

CAFA-IVR v1.0

Counterfactual ASR Failure Attribution for Conversational IVR

Reference Specification — Vijay Kumar Sridharan — August 2026
IEEE Senior Member — Frisco, TX, USA

1. Purpose

CAFA-IVR defines a portable test procedure for determining whether adding an audio/ASR layer causes a conversational test to fail when the same semantic request succeeds through a reference-text control. The framework is independent of ASR vendor, NLU vendor, contact-center platform, and dialogue architecture.

2. Required paired execution

For every semantic test i , an implementation **MUST** execute both paths using the same frozen downstream configuration:

$$T_i : \text{reference text} \rightarrow \text{downstream system}, \quad A_i : \text{audio} \rightarrow \text{ASR} \rightarrow \text{downstream system}.$$

The two paths **MUST** share the same expected semantic outcome. The downstream component may be an intent classifier, workflow, LLM agent, or tool-using system, provided a reproducible task oracle exists.

3. Attribution states

Text	Audio	State	Interpretation
Pass	Pass	HEALTHY	End-to-end semantic success.
Pass	Fail	SPEECH_ATTRIBUTABLE	Failure appears only after the speech path is introduced.
Fail	Fail	DOWNSTREAM_OR_TEST	Reference text already fails; do not attribute the failure to ASR.
Fail	Pass	CONTEXT_OR_ORACLE	Investigate nondeterminism, context, or the expected-result oracle.

A speech-attributable state narrows the failure surface; it does not by itself prove one acoustic root cause.

4. Metrics

WER. Preserve conventional lexical fidelity:

$$\text{WER} = \frac{S + D + I}{N}.$$

ASR-IFR. Let $F_i = 1$ when the text control passes and the audio-path task fails, else 0:

$$\text{ASR-IFR} = \frac{1}{N} \sum_i F_i.$$

A conditional variant may divide by the number of text-control successes.

CEER. Critical-Entity Error Rate measures governed semantic values such as amounts, dates, negation, account type, authentication values, and lock/unlock/cancel/confirm actions. Entity values **SHOULD** be normalized before scoring. For a result set, CEER is the total number of missing and spurious critical entities divided by the total number of reference critical entities. It is an unbounded normalized error count, not a probability, and may exceed 1.0 when spurious entities outnumber reference entities.

CIER (optional). CIER measures realized consequence. Let $\mathcal{I}$ be the set of trials carrying a recognized impact level, let $A_i = 1$ when the audio-path task succeeds and 0 when it fails, and let $w(L_i)$ be the selected impact-level weight. Then

$$\text{CIER} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (1 - A_i) w(L_i).$$

The reference scorer uses illustrative weights L0=0, L1=0, L2=.25, L3=.65, L4=1.0. A successful audio-path task contributes zero rather than being excluded, so CIER is a severity-weighted failure rate over the impact-labelled population, not an average severity of failures. A failed task contributes its L0–L4 weight; a trial without a recognized impact level is excluded. CIER counts realized audio-path task failures regardless of attribution. An ASR-attributable consequence analysis should restrict the population to **SPEECH_ATTRIBUTABLE** trials before summarizing. The weights are policy examples, not universal thresholds.

5. Minimum trial record

Field	Status	Purpose
<code>test_id</code>	Required	Stable semantic test identifier.
<code>condition</code>	Required	Clean, telephony, noise, rate, or another controlled condition.
<code>reference_text</code>	Required	Canonical human-authored reference.
<code>asr_transcript</code>	Required	Observed speech-engine transcript.
Expected/observed outcome	Required	Intent labels or equivalent task oracle.
<code>text_control_pass</code>	Required	Downstream result when ASR is bypassed.
<code>audio_task_success</code>	Required	Outcome through audio $\rightarrow$ ASR.
Entity JSON	Recommended	Governed critical values and observed values.
<code>risk_tier</code>	Recommended	Local Low/Medium/High or equivalent.
Versions	Recommended	ASR, NLU/agent, prompt/tool/oracle configuration.

6. Minimum acoustic matrix

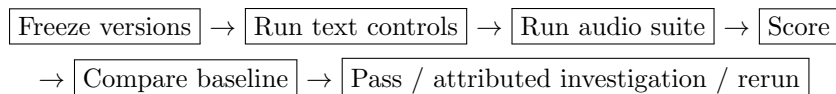
A production suite **SHOULD** establish clean/wideband baseline, 8-kHz or actual telephony path, moderate and severe controlled noise, bounded speaking-rate variation, representative human speakers, and critical-semantic cases. Synthetic speech may expand coverage but should not be the only evidence used for a human-robustness claim.

7. Release-gating policy

CAFA-IVR does not prescribe universal numeric thresholds. A team should compare a candidate against its last approved baseline and use stricter policy for high-consequence journeys. A practical policy is:

1. no unexplained meaningful increase in ASR-IFR;
2. no new governed critical-entity failures in high-risk journeys without explicit approval;
3. WER regressions are investigated, but WER alone does not decide the release;
4. text-control failures are routed to NLU/agent/test owners rather than ASR owners;
5. failed gates retain audio, transcript, versions, expected outcome, and observed outcome for replay.

8. CI/CD lifecycle



An adapter only needs to populate the trial record. Credentials should stay in the adopting platform’s standard credential mechanism and must not be committed to a CAFA results package.

9. Reproducibility and independent adoption

An auditable run should freeze the audio or source identifiers, perturbation parameters and random seeds, ASR version, downstream version, prompts/tools when applicable, locale, endpoint configuration, test-oracle version, and CAFA-IVR specification version.

The v1.0 reference release recomputes scores from supplied decoded outputs. It does not include the audio-generation, ASR-decoding, perturbation, or bootstrap-resampling pipelines and therefore does not reproduce the reported decode-level and bootstrap results end to end. The included 360-trial pilot contains no critical-entity annotations or impact labels: it exercises WER and ASR-IFR, but does not empirically exercise CEER or CIER.

An organization may accurately state that it adopted CAFA-IVR only when it actually used the paired procedure. A useful implementation statement is:

“We evaluated conversational voice regressions using CAFA-IVR v1.0 paired reference-text and audio-path testing and incorporated ASR-IFR into our failure-triage workflow.”

Independent adopters should retain test scope, decision, resulting engineering change, and artifacts sufficient for a third party to verify the use.

10. Conformance checklist

A CAFA-IVR v1.0 conforming implementation should be able to answer *yes* to the following:

- Each semantic test has a stable ID, reference text, and expected outcome.
- Reference text and audio use the same frozen downstream configuration.
- The ASR transcript and both path outcomes are retained at trial level.
- ASR-IFR is calculated from the paired outcomes, not inferred from WER.
- High-risk critical entities are explicitly annotated or the absence of CEER is documented.
- Candidate results are compared with a named approved baseline when used for release gating.
- Failed trials retain enough evidence for replay and independent review.
- The CAFA-IVR specification version and tested ASR/downstream versions are recorded.

11. Independent adoption record

When the framework is evaluated or adopted by an independent team, retain the following fields so use can be verified later:

Field	Record
Organization / team	Name and business unit
Evaluator	Name, title, and contact
CAFA-IVR version	Specification/release used
Technical scope	ASR, NLU/agent, number of tests, number of audio trials
Evidence type	Human, synthetic, or mixed speech; acoustic conditions
Operational use	Test plan, CI job, dashboard, runbook, release decision, or evaluation only
Engineering impact	Configuration/model change, avoided misdiagnosis, new gate, issue found, release decision
Evidence retained	Report, screenshot, commit, run ID, URL, artifact hash, or internal document reference
Public citation permission	Yes / No / Limited

12. Non-endorsement

CAFA-IVR is a technical reference framework. Listing an employer, vendor, dataset, or platform does not imply sponsorship or endorsement. Production thresholds and risk decisions remain the responsibility of the adopting organization.